SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	BTECH-AIML/2024	**Date:**	

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.
Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images. **Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.**

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	**Course Coordinator**	**Program Director**

Q1. Improving Night-Time CCTV Images Using Digital Image Processing

Introduction

In shopping malls, CCTV cameras are used for security and monitoring. At night, captured images may have low brightness, sensor noise, poor contrast, and unclear object boundaries. Digital image processing techniques can improve these images before applying object detection algorithms.

1. Image Acquisition

The CCTV camera captures images of people, vehicles, and objects in the mall. These images are used as the input for the image enhancement process.

2. Noise Reduction Using Filters

Low-light images often contain unwanted noise. Gaussian filtering reduces random noise, while median filtering removes isolated noisy pixels. Bilateral filtering can reduce noise while preserving important edges.

3. Brightness and Contrast Enhancement

Brightness adjustment increases the visibility of dark regions. Histogram equalization improves the overall distribution of intensity values, while CLAHE enhances local contrast without excessively increasing contrast across the entire image.

4. Image Sharpening

Unsharp masking and Laplacian filtering enhance edges and fine details. These techniques help make object boundaries clearer, but excessive sharpening may amplify noise.

5. Object Detection

The enhanced image is supplied to a detection model such as YOLO to locate people, bags, and other objects. The detected objects can then be tracked across video frames.

Summary of Techniques

Stage	Purpose	Common Methods
Image Acquisition	Capture input images from CCTV	CCTV camera
Noise Reduction	Remove low-light and sensor noise	Gaussian, median, bilateral filters
Brightness and Contrast	Make dark regions more visible	Histogram equalization, CLAHE
Sharpening	Make object boundaries clearer	Unsharp masking, Laplacian filter
Object Detection	Locate people, bags and objects	Detection model such as YOLO, tracking

Processing Sequence

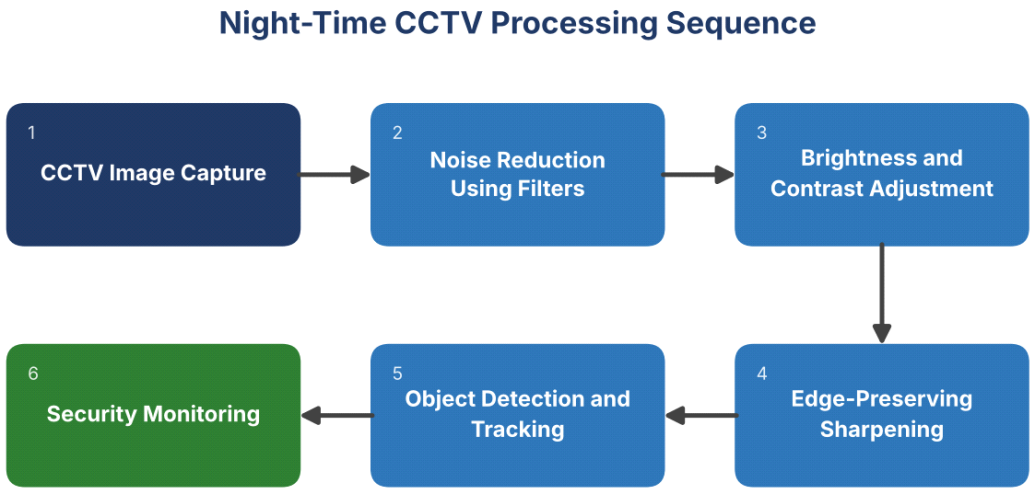


Figure 1: Processing sequence for night-time CCTV images

Advantages

- Improves visibility in low-light conditions.
- Reduces image noise and enhances details.
- Helps detect objects in CCTV footage.
- Supports security monitoring and suspicious-activity detection.

Conclusion

Combining noise reduction, contrast enhancement, and sharpening improves night-time CCTV images. These operations help object detection systems perform more effectively, although extremely dark or blurred images may still lack sufficient information.

Q2. Improving Document Images Using Image Processing Techniques

Introduction

A university document-scanning system must convert photographed documents into clear digital images. Shadows, noise, uneven lighting, page rotation, and blurred text can reduce readability. Image processing helps correct these problems before text recognition.

1. Image Filtering

Median filtering removes isolated noise, and Gaussian filtering smooths unwanted variations. These filters must be applied carefully to preserve thin character strokes.

2. Illumination and Contrast Correction

Uneven illumination can make one part of a page brighter than another. Background estimation and illumination normalization can reduce these variations. CLAHE improves local contrast, making faded text easier to read.

3. Thresholding

Thresholding separates text from the paper background. Global thresholding uses a single threshold for the image, while adaptive thresholding calculates thresholds for local regions. Adaptive thresholding is more appropriate when the document contains shadows or uneven brightness.

4. Edge Detection

Canny or Sobel edge detection identifies the boundaries of the page and the edges of printed content. The detected page boundary can be used to estimate the page's orientation and perspective.

5. Geometric Correction and Segmentation

Perspective correction transforms a tilted document into a rectangular view. Skew correction straightens rotated text. Segmentation then separates text lines, paragraphs, and individual characters or words.

6. OCR Processing

Optical Character Recognition (OCR) converts the processed document image into machine-readable text. The resulting text can be searched, edited, and stored digitally.

Role of Each Technique in the Pipeline

Technique	Problem Addressed	Role in the Pipeline
Filtering	Noise, small variations	Cleans the image while preserving thin strokes
Illumination and Contrast Correction	Uneven lighting, faded text	Normalizes brightness and improves local contrast
Thresholding	Shadows, paper background	Separates text from the background
Edge Detection	Unknown page boundary	Finds page edges to estimate orientation
Geometric Correction	Tilt and perspective	Straightens and rectangles the

		page
Segmentation and OCR	Unstructured layout	Separates text regions and converts them to digital text

Processing Pipeline

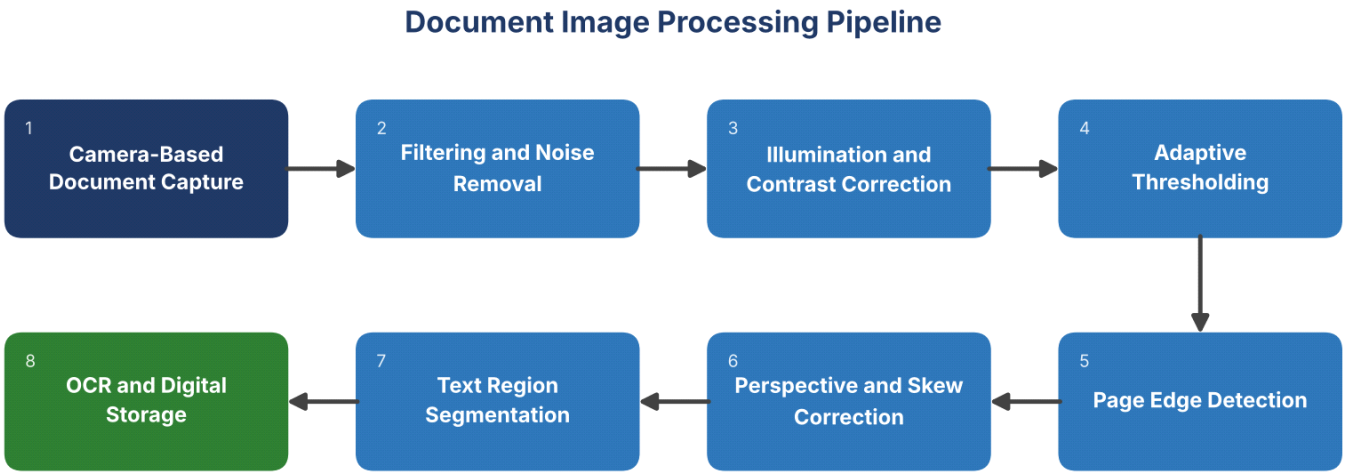


Figure 2: Processing pipeline for document images

Advantages

- Produces clear digital documents.
- Improves readability of faded text.
- Corrects tilted and perspective-distorted pages.
- Supports automatic text extraction and digital archiving.

Conclusion

Filtering, illumination correction, thresholding, edge detection, and segmentation improve document quality. Combining these techniques with geometric correction and OCR creates an efficient document digitization system.

Q3. Comparing Image Segmentation Methods for Apple Defect Detection

Introduction

Fruit-processing industries require automatic inspection to identify damaged apples before packaging. Image quality may be affected by shadows, reflections, varying backgrounds, and differences in fruit color. Image segmentation separates the apple and its defective regions for analysis.

1. Global Thresholding

Global thresholding separates pixels using a single threshold value.

- Simple and computationally efficient.
- Works well under uniform lighting and against a consistent background.
- May fail when healthy and damaged areas have similar intensities or when shadows change pixel brightness.

2. Adaptive Thresholding

Adaptive thresholding calculates a threshold for each local image region.

- Handles uneven lighting better than global thresholding.
- Can highlight dark spots and local discoloration.
- May mistakenly classify natural skin texture or shadows as defects.

3. Edge-Based Segmentation

Edge-based methods detect changes in intensity using operators such as Canny and Sobel.

- Useful for identifying apple boundaries and visible cracks.
- Helps separate regions with distinct boundaries.
- May fail when defects have weak edges or reflections produce false boundaries.

4. Region-Based Segmentation

Region-based techniques group neighboring pixels according to similar color, intensity, or texture.

- Useful for identifying connected patches of damaged tissue.
- Can analyze the area and shape of defects.
- May merge healthy and damaged regions when their visual characteristics are similar.

Comparison Table

Method	Best Application	Major Drawback
Global thresholding	Uniform backgrounds and lighting	Sensitive to illumination
Adaptive thresholding	Local brightness variations	Sensitive to local texture
Edge-based segmentation	Cracks and clear boundaries	Weak edges may be missed
Region-based segmentation	Connected discoloration patches	Similar regions may merge

Proposed Solution

A combined segmentation method can improve the system:

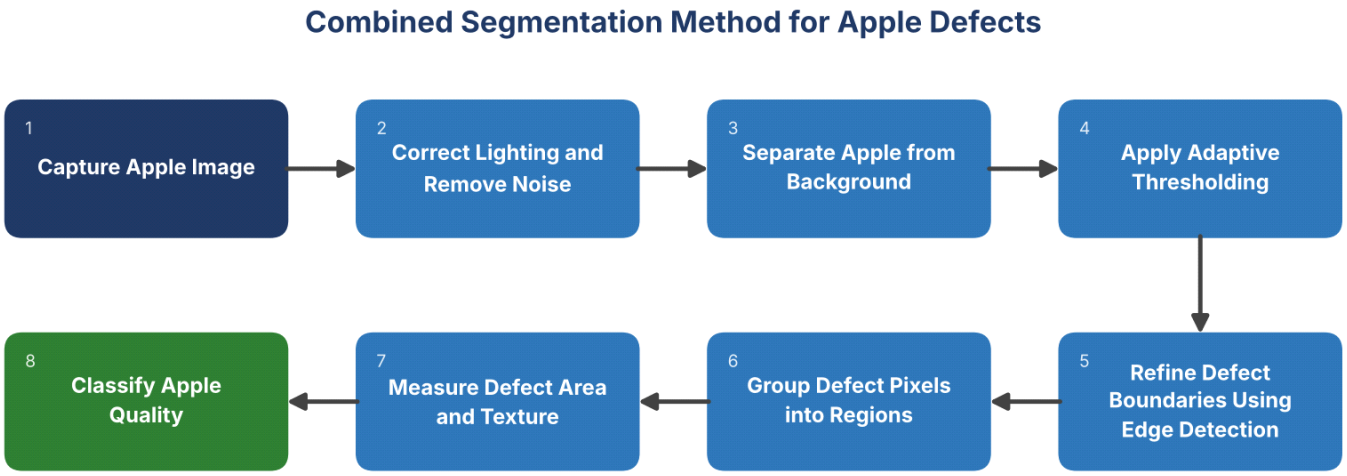


Figure 3: Combined segmentation method for apple defect detection

Adaptive thresholding detects local dark regions, edge detection refines their boundaries, and region-based analysis measures the size and shape of suspected defects. A trained classifier can use these measurements along with color and texture to determine whether an apple is healthy or damaged.

Advantages

- Detects visible defects automatically.
- Reduces manual inspection time.
- Improves fruit quality assessment.
- Supports automatic grading and sorting.

Conclusion

Global thresholding is suitable for controlled conditions, while adaptive thresholding is more useful under varying illumination. Combining thresholding,

edge detection, and region analysis provides a more flexible approach to apple defect detection.

Q4. Image Processing for Reliable Lane Detection

Introduction

Autonomous vehicles use camera images to identify lane markings and maintain their position on the road. Shadows, rain, faded markings, vehicles, and changing illumination can affect detection. A sequence of image processing operations helps extract lane information from road images.

1. Grayscale Conversion

The color image is converted into grayscale to simplify processing and reduce computational requirements.

Limitation: Some differences between yellow lane markings and the road may be lost. Color information can be retained for additional detection.

2. Noise Filtering

Gaussian filtering reduces noise caused by the camera and weather conditions.

Limitation: Excessive smoothing may remove thin lane markings and small road details.

3. Contrast Enhancement

CLAHE improves local contrast between lane markings and road surfaces, especially in shadowed areas.

Limitation: It may also enhance road texture, shadows, and unwanted image features.

4. Edge Detection

Canny edge detection identifies boundaries where image intensity changes sharply. These edges can represent lane markings and road boundaries.

Limitation: Vehicle boundaries, cracks, and road shadows can also produce strong edges.

5. Thresholding

Intensity or color thresholding selects pixels likely to belong to lane markings. White and yellow lane markings can be identified using suitable intensity and color ranges.

Limitation: Rain, glare, worn paint, and nighttime conditions can make fixed thresholds unreliable.

6. Region of Interest

A polygon-shaped ROI is applied to focus on the road ahead, reducing irrelevant information from buildings, the sky, and roadside objects.

Limitation: A fixed ROI may not work on sharp turns, steep roads, or different camera positions.

7. Lane Detection and Tracking

Hough line detection or curve-fitting methods identify lane boundaries from the processed image. Tracking across consecutive frames helps maintain stable lane estimates.

Stage-wise Limitations at a Glance

Stage	Purpose	Main Limitation
Grayscale Conversion	Simplify processing	Yellow marking contrast may be lost
Noise Filtering	Reduce camera and	Over-smoothing removes thin

	weather noise	markings
Contrast Enhancement	Improve marking visibility	May enhance texture and shadows
Edge Detection	Find lane boundaries	Vehicle edges, cracks, shadows give false edges
Thresholding	Select lane marking pixels	Rain, glare, worn paint, night conditions
Region of Interest	Focus on the road ahead	Fixed ROI fails on turns, slopes, camera shifts
Detection and Tracking	Fit and follow lane boundaries	Needs stable tracking across frames

Processing Pipeline

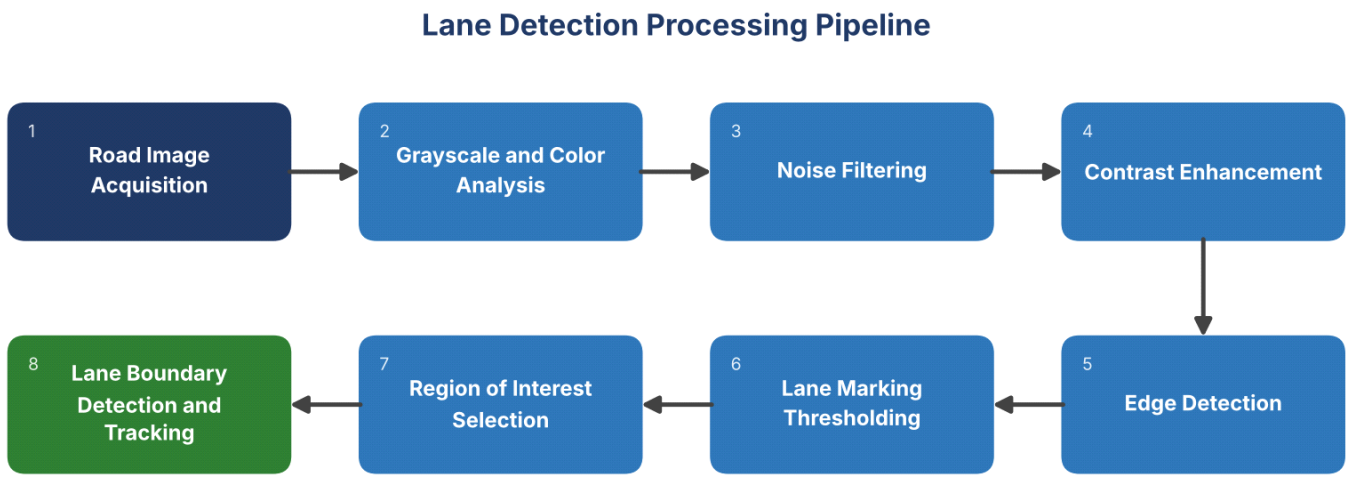


Figure 4: Lane detection processing pipeline

Advantages

- Helps identify lane markings.
- Reduces irrelevant image information.
- Supports lane departure warning and lane keeping.
- Enables real-time analysis of road images.

Conclusion

Grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and ROI selection provide a foundation for lane detection. Combining these methods with temporal tracking and robust detection models helps address changing weather and lighting conditions.

Q5. Automated Waste Classification Using Computer Vision

Introduction

Municipal corporations can use computer vision to classify waste into plastic, paper, and metal. Images captured above conveyor belts may contain overlapping objects, shadows, reflections, noise, and different orientations. A complete image-processing pipeline can extract useful information and classify each waste item.

1. Image Acquisition

A camera captures images of waste moving along a conveyor belt. Consistent camera positioning and suitable lighting improve the quality of the captured images.

2. Preprocessing

Preprocessing includes image resizing, noise removal, and correction of unwanted variations. Median filtering removes isolated noise, while Gaussian filtering reduces small intensity fluctuations.

3. Image Enhancement

Contrast enhancement improves object visibility. CLAHE can improve local contrast, and illumination normalization helps reduce differences caused by shadows and brightness changes. Color information should be preserved because it may help distinguish different waste materials.

4. Segmentation

Segmentation separates waste objects from the conveyor background. Thresholding, contour detection, and background subtraction can be used when objects are separated. When objects overlap or touch, watershed segmentation or instance-segmentation models can help separate individual items.

5. Feature Extraction

Useful characteristics are extracted from the segmented objects:

Feature	What It Describes
Color features	Identify differences in surface color
Shape features	Describe object geometry and outline
Texture features	Distinguish smooth, fibrous, and rough surfaces
Edge features	Represent boundaries and structural details
Size features	Estimate the area and dimensions of each object

6. Classification

A trained model classifies waste using the extracted features. SVM can be used with manually designed features, while CNNs can learn visual characteristics from labeled training images. The output category is plastic, paper, or metal. Objects that cannot be classified confidently can be sent for additional inspection.

7. Automated Sorting

The classification result is sent to a conveyor controller. A robotic arm, air jet, or mechanical diverter directs each object to its appropriate recycling bin.

Complete Pipeline

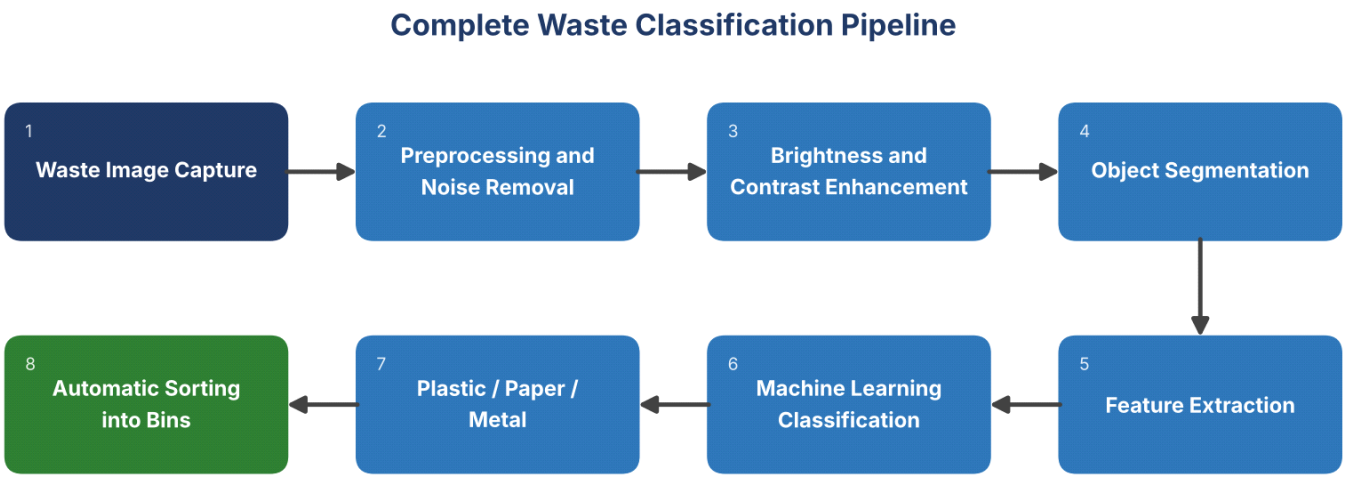


Figure 5: Complete waste classification pipeline

Handling Practical Challenges

Challenge	Solution
Shadows	Illumination correction and CLAHE
Reflections	Diffused lighting and color normalization
Noise	Median or Gaussian filtering
Overlapping objects	Watershed or instance segmentation
Different orientations	Rotation-invariant features or CNNs
Similar-looking materials	Texture analysis and a diverse training dataset

Advantages

- Reduces manual sorting.
- Improves sorting speed and consistency.
- Supports efficient recycling.
- Reduces mixing of different waste materials.
- Can be integrated into smart waste-management systems.

Conclusion

An automated waste-classification system combines preprocessing, enhancement, segmentation, feature extraction, and classification. Appropriate algorithms and controlled lighting help handle real-world variations and improve the reliability of plastic, paper, and metal sorting.